

Cluster 83853

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Subjects

Research fields: Engineering, Computer science, Mathematics

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Acoustics, Mechanics, Petroleum engineering

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: Mental Health, Flow Measurement, distributed acoustic sensing, phase flow rates, LWD

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

1.30% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

1.30% of English-language articles are related to cybersecurity

Vital signs

78 recent articles in the cluster

78 articles in this cluster were published in the five year period ending in 2023.

Average article is **22.07** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **84.69th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 84.69% of other clusters.

Citation rating: **41.11th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 41.11% of all other articles published in the same year.

89.74% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 60 of 78 articles. 51 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	18	0.65
United States	15	2.09
United Kingdom	9	0.42
Norway	3	0.71
India	3	1.40
Saudi Arabia	2	1.50
United Arab Emirates	2	0.50
Russia	2	0.13
Switzerland	1	3.00

Top funding countries

Data available for 4 of 78 articles. 4 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	4	1.33

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Flow Measurement Optimization Using Surface Measurements and Downhole Sound Speed Measurements from Local or Distributed Acoustic Sensors](#)

2021: SPE Production & Operations. 5 citations.

[Innovative Workflow for Wellbore Stability Evaluation Using Logging-While-Drilling Technologies, 2023, Day 3 Tue](#)

2023: 2023. 1 citations.

[The Mining Method of Ideological and Political Elements in University Public Mental Health Courses Based on Artificial Intelligence Technology](#)

2022: Journal of environmental and public health. 7 citations.

[Social Determinants of Mental Health: Recommendations for Research, Training, Practice, and Policy.](#)

2022: JAMA psychiatry. 46 citations.

[EVALUATION OF LWD HIGH RESOLUTION ULTRASONIC IMAGING TECHNOLOGY AND APPLICATIONS IN SLIM HOLE SIZE](#)

2021: SPWLA 62nd Annual Online Symposium Transactions. 5 citations.

[Social determinants of mental health and adolescent anxiety and depression: Findings from the 2018 to 2019 National Survey of Children's Health](#)

2022: The International journal of social psychiatry. 9 citations.

[Repeat DAS and DTS Production Logs on a Permanent Fiber Optic Cable for Evaluating Production Changes and Interference with Offset Wells](#)

2022: Proceedings of the 10th Unconventional Resources Technology Conference. 7 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

No data available.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[The Mining Method of Ideological and Political Elements in University Public Mental Health Courses Based on Artificial Intelligence Technology](#)

2022: Journal of environmental and public health. 7 citations.

[The Singular Value Decomposition](#)

2021: Practical Linear Algebra. 23 citations.

[Critical Periods in Child Development and the Transition to Adulthood.](#)

2021: JAMA network open. 13 citations.

[Social Determinants of Mental Health: Recommendations for Research, Training, Practice, and Policy.](#)

2022: JAMA psychiatry. 46 citations.

[EVALUATION OF LWD HIGH RESOLUTION ULTRASONIC IMAGING TECHNOLOGY AND APPLICATIONS IN SLIM HOLE SIZE](#)

2021: SPWLA 62nd Annual Online Symposium Transactions. 5 citations.

[Real-Time 3D Anisotropy Analysis Enables Lithology Identification at Distance](#)

2022: SPWLA 63rd Annual Symposium Transactions. 14 citations.

[Social determinants of mental health and adolescent anxiety and depression: Findings from the 2018 to 2019 National Survey of Children's Health](#)

2022: The International journal of social psychiatry. 9 citations.

[Repeat DAS and DTS Production Logs on a Permanent Fiber Optic Cable for Evaluating Production Changes and Interference with Offset Wells](#)

2022: Proceedings of the 10th Unconventional Resources Technology Conference. 7 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 77 out of 78 articles. 15 of these articles came from these sources.

Source	Articles (last five years)
SPE Annual Technical Conference and Exhibition	3
Journal of environmental and public health	2
The Random Spatial Economy and Its Evolution	2
Sensors (Basel, Switzerland)	2
Journal of Petroleum Technology	1
Day 2 Tue, September 01, 2020	1
arXiv: Classical Physics	1
Flow Measurement and Instrumentation	1
REVISTA DE PSICOLOGIA DEL DEPORTE	1
Reslaj Religion Education Social Laa Roiba Journal	1

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 78 of 78 articles. 18 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Ahmed Taher	7	1.68
Ö. Haldun Ünalmiş	4	0.82
A. Taher	3	0.33
Mohamed Fouda	3	2.25
M. Fouda	3	0.33
田瑞峰	2	0.00
Ahmed Al Kiyoumi	2	0.83
Dilip V Jeste	2	25.33
Shardul Parihar	2	0.83
Erick Jimenez	2	1.83

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 67 of 78 articles. 20 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Halliburton (United Kingdom)	6	0.53
Weatherford (Norway)	3	0.71
Saudi Aramco (United States)	3	0.75
Abu Dhabi National Oil (United Arab Emirates)	2	0.5
哈尔滨工程大学核科学与技术学院 - China	2	0
Saudi Aramco (Saudi Arabia)	2	1.5
University of California, San Diego - United States	2	8.83
Naval Research Institute, 100161 Beijing, China	1	0
Ecopetrol (Colombia)	1	0.4
Č VUT v Praze, Fakulta strojní, Ústav technické matematiky, Karlovo náměstí 13, 121 35 Praha 2, Česká republika	1	1

Organization types

This table covers the 53 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 78 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
20.30%	36.09%	7.52%	0.00%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 20 out of 78 articles. 17 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Halliburton (United Kingdom)	6	0.53
Weatherford (Norway)	3	0.71
Saudi Aramco (United States)	3	0.75
Saudi Aramco (Saudi Arabia)	2	1.5
Abu Dhabi National Oil (United Arab Emirates)	2	0.5
China National Offshore Oil Corporation (China)	1	1
China National Petroleum Corporation (China)	1	0.2
Eni (Italy)	1	0
Total (United Kingdom)	1	0
Ecopetrol (Colombia)	1	0.4

Top funders

Data available for 8 out of 78 recent articles. 8 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	4	1.33
Research results of the Beijing Higher Education Institute 2022 Topics "Research on the Reform of Practical Teaching of Labor and Social Security Law'Course under OBE Concept"	1	0.00
Open Foundation of Key Laboratory of Thermal Power Technology	1	2.00
Shaanxi Province - China	1	0.33
68th Batch of China Postdoctoral Foundation	1	0.33
The authors would like to thank the Southwest Research Institute (SwRI)'s Multiphase Flow Facility's operators/engineers for their help with the experimental setup and data collection.	1	1.00
Southwest Research Institute	1	1.00
Fundamental Research Funds for the Central Universities- Doctoral Research Innovation Fund Project	1	2.00
Weatherford	1	1.25

Funder types

This table covers the 4 articles in this cluster from the last five years with data about their funding organization types (out of 78 articles total).

Commercial	Education	Nonprofit	Government
0.00%	0.00%	0.00%	100.00%

Patents and industry

64 patents cite articles in this cluster (**72.7th** percentile among all clusters)

7.83% of articles in the cluster are cited by patents

15.88% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
ANNULUS MUD FLOW RATE MEASUREMENT WHILE DRILLING AND USE THEREOF TO DETECT WELL DYSFUNCTION	6
DETECTING A KICK IN A WELLBORE	2
DOWNHOLE ACOUSTIC DENSITY DETECTION	2
DOWNHOLE CALIPER USING MULTIPLE ACOUSTIC TRANSDUCERS	2
SUBSURFACE FIBER OPTIC STIMULATION-FLOW METER	2
SYSTEMS AND METHODS FOR DYNAMIC REAL-TIME WATER-CUT MONITORING	2
DOWNHOLE 3-PHASE FLOW MEASUREMENT USING SPEED OF SOUND ABOVE AND BELOW THE BUBBLE-POINT PRESSURE	2
FLUID INFLOW CHARACTERIZATION USING HYBRID DAS/DTS MEASUREMENTS	1
MULTIPHASE FLOW MEASUREMENT	1

Top patent assignees

Patent assignee	Same-cluster citations
BP (UNITED KINGDOM)	7
HAITHAM RAKAD MOHAMMAD AL HATAMLEH	7
SAUDI ARABIAN OIL COMPANY	6
ALBERTY MARK W	6
BAKSALT GMBH	6
BP CORPORATION NORTH AMERICA INC	6
BP CORPORATION NORTH AMERICA INC.	6
SATURN LICENSING, LLC	5
SCHLUMBERGER TECHNOLOGY CORPORATION	5

Connections

Top citing clusters

No data available.

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
64465	1	Auckland Council, Waiheke Local Board, Waiheke community, Council economic development, Local Board Plan
14679	1	Particle size distribution, Particle size, ultrasonic attenuation, size distribution measurement, attenuation coefficient
33089	1	distributed acoustic sensing, DAS data, acoustic sensing data, DAS Seismic Data, DAS strain measurements
15086	1	Patient Health Questionnaire, Generalized Anxiety Disorder, Patient Health, Anxiety Disorder scale, mental health outcomes
38397	1	Mental Health Chatbots, health care, Artificial Intelligence, Conversational Agents, digital mental health
7955	1	Discontinuous Galerkin Method, compressible two-phase flows, two-phase flow model, Riemann Problem, numerical
25645	1	pressure waves, acoustic waves, gas bubbles, wave propagation, bubble liquid
7340	1	gas flow, Knudsen number, lattice Boltzmann method, rarefied gas flow, Heat Transfer
17513	1	LWD resistivity measurements, electromagnetic resistivity LWD, Ultra-Deep Resistivity Inversions, LWD azimuthal electromagnetic, Deep Azimuthal Resistivity
811	1	mental health, University Health System, Health Care Workers, National University Health, mental health consequences

Top GitHub repositories

No data available.

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).